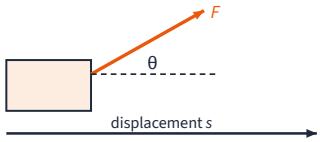
CHAPTER 4

WORK, ENERGY & POWER

Key idea: work transfers energy; energy is never created or destroyed, only converted.

1 WORK DONE



$$W = Fs\cos\theta$$

- Scalar, unit J (1 J = 1 N × 1 m)
- Variable force: $W = \int F\,dx$ = **area** under F–s
- $\theta = 90^\circ$ (centripetal force, normal reaction): $W = 0$
- Friction does negative work

2 ENERGY STORES

$$E_k = \tfrac{1}{2}mv^2 \qquad \Delta U = mg\Delta h$$

$$U_{\text{spring}} = \tfrac{1}{2}kx^2$$

Elastic energy = area under the F–x graph.

3 POWER

$$P = \frac{W}{t} = \frac{dW}{dt} \qquad P = Fv$$

- Rate of doing work, unit W = J s^{-1}
- Constant speed against resistance F: $P = Fv$
- 1 kW h = 3.6 × 10^6 J

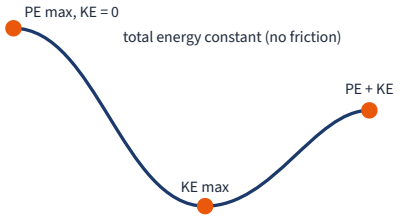
4 WORK–ENERGY THEOREM

$$W_{\text{net}} = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2$$

Net work done on a body = change in its kinetic energy.

Against friction on a level surface: $W = \mu mgs$.

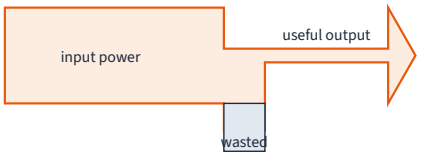
5 CONSERVATION OF ENERGY



$$\tfrac{1}{2}mu^2 + mgh_1 = \tfrac{1}{2}mv^2 + mgh_2 + F_f d$$

- No friction: mechanical energy constant
- From rest through height h: $v = \sqrt{2gh}$, any smooth path
- Pendulum: $v = \sqrt{2gL(1-\cos\theta)}$

6 EFFICIENCY



$$\eta = \frac{\text{useful output}}{\text{input}} \times 100\%$$

The rest is wasted, mostly as heat; $\eta < 100\%$.

7 CONSERVATIVE OR NOT?

Conservative	work independent of path (gravity, spring); energy recoverable
Non-conservative	work depends on path (friction, air resistance); energy → heat

Lifting through height h: $W = mgh$ by any route.

8 COMMON SECTION A TRAPS

- Using the full force instead of its component along s.
- Taking $P = Fv$ with the **driving** force when speed changes; use the net force only for ΔE_k.
- Forgetting the friction term in an energy equation.
- Mixing kW h and J.

EXAM TIPS

- ✓Write "loss of PE = gain in KE + work against friction".
- ✓Use energy methods when the path is curved.
- ✓Use $P = Fv$ for vehicles at constant speed.
- ✓Give efficiency as a percentage or a fraction, consistently.